Name: ____

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The First Spell Checker

The very first spell checker would use misspelled words as inputs. For misspelled each word, it would:

(1) Develop a list of *alternative words* by **deleting**, **substituting**, or **inserting** at each position within the misspelled word, or **swapping** the positions of two adjacent letters.

(2) Search its dictionary of correct spellings to see which of the alternatives on the list were real words, and throw away the other "nonsense" spellings.

(3) Produce a list of real, correctly spelled words for the user to choose from.

Misspelled Word: `wello`

Deleting

1) For each of the five letters in "wello", write an alternative spelling with one character deleted. *We've done the first one to get you started.*

ello	______	______	______	______
delete 1st letter	delete 2nd letter	delete 3rd letter	delete 4th letter	delete 5th letter

★ How many alternate "spellings" can be created from *every* possible deletion? ____________________

Substituting

2) For each of the five letters in "wello", replace them with a different one. *We've provided a first option to get you started.*

dello	______	______	______	______
replace 1st letter	replace 2nd letter	replace 3rd letter	replace 4th letter	replace 5th letter

3) How many possible substitutions *could* you make for the 1st letter? ____________________

★ How many alternate "spelling" can be created from *every* possible substitution? ____________________

Inserting

4) For each of the five letters in "wello", write an alternative "spelling" with one character inserted before each position. *We've done the first one to get you started.*

zwello	______	______	______	______
before 1st letter	before 2nd letter	before 3rd letter	before 4th letter	before 5th letter

5) How many possible insertions *could* you make after the 1st letter? ____________________

★ How many alternate "spelling" can be created from *every* possible insertion? ____________________

Swapping

6) Swap the positions of each pair of **adjacent** letters. *We've done the first two to get you started.*

ewllo	______	______	______
1st pair (WE)	2nd pair (EL)	3rd pair (LL)	4th pair (LO)

7) How many possible swaps could you make for this "spelling"? ______________

Reflect

8) How many real "spelling" did you come up with? ________ What were they? ____________________

9) Compared to your own strategy for spell-checking, how similar / different is the first spell checker's algorithm?

____

____

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A Pyret Spell Checker

Open the [Spell Checker Starter File](#) and click "Run". Follow each of the instructions below to discover how our very own Pyret spell checker works.

Suggestions for 1 Edit-Distance

`deletions`, `insertions`, `swaps`, and `subs` consume a word, and suggest a list of words that can be reached by making a **single change**.

1) Fill in the 2nd column of table below by evaluating each expression, and recording the **number of suggestions** Pyret returns.

Expression	# Suggestions	Did Pyret find all your suggestions?
`deletions("wello")`		**Yes** **No**, it missed:
`insertions("wello")`		**Yes** **No**, it missed:
`swaps("wello")`		**Yes** **No**, it missed:
`subs("wello")`		**Yes** **No**, it missed:

2) **Flip back to [The First Spell Checker](#).** How many of YOUR suggestions are in Pyret's list? Fill in the 3rd column of the table above.

Suggestions for Larger Edit-Distances

3) Evaluate `swaps(swaps("wello"))`. What does it do? ______________________________

____

4) Evaluate `swaps(subs("wello"))`. What does it do?: ______________________________

____

Removing "Junk" Words

Pyret has a function `only-real`, that takes in a table of suggested words and a dictionary.

5) Evaluate `insertions("flt")`. How many suggestions does it return? ______________________________

6) Evaluate the two expressions below and record the **number of words** they return

`only-real(insertions("pat"), WORDS-S)`	
`only-real(insertions("pat"), WORDS-M)`	
`only-real(insertions("pat"), WORDS-L)`	

7) What does `only-real` do? Why do we get different results? ______________________________

____

Putting it All Together

The `alt-words` function combines all 5 of these functions. It takes in a "word", a dictionary, and a maximum edit-distance.

8) If you had to guess, what word did the author mean to type when they wrote `crosswrd`? ______________________________

9) Evaluate `alt-words("crosswrd", WORDS-L, 2)`. Record the first 3 results. Describe the edit(s) to get here from "crosswrd".

Alternative Word	# Edits	Edits from "crosswrd"

10) If the spellchecker were to take the number of edits into consideration, which of these suggestions do you think it would propose first?

Why? ______________________________

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